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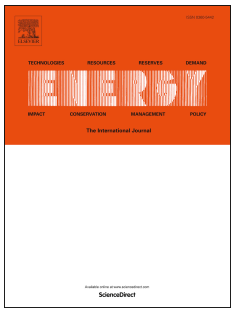
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Md. Mashiur Rahman	Author performed the section of the construction, commissioning, test, characterization, and modelling.
Ulrik Birk Henriksen	Author performed the section of the design, construction, test, and commissioning. All work was supervised by the author.
Jesper Ahrenfeldt	Author performed the section of the design, construction, test, and commissioning.
Maria Puig Arnavat	Author performed the section of the commissioning, test, and modelling.

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Design, Construction and Operation of a Low-Tar Biomass (LTB) Gasifier for Rural Applications

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Highlights:

- LTB Gasifier produces an almost tar free producer gas.
- Biomass with 9-34% MC results in very low tar contents (10.6 – 29.8 mg/m^3).
- Producer gas from LTB gasifier can be used directly in internal combustion engine.
- Temperature profile of the LTB gasifier observed relatively low.
- LTB gasifier performance can be estimated using a thermodynamic equilibrium model.

Abstract

A low-tar biomass (LTB) gasifier with internal recirculation of pyrolysis gas and a separate combustor installed in the partial oxidation zone has been designed, developed and operated. The gasifier mixes the pyrolysis gases with gasifying air and burns the mixture in a combustor while maintaining a very low tar content in the producer gas that makes it suitable for direct use in an internal combustion engine. The resulting tar content is much lower than those obtained in previous studies for the same type of moving bed gasifier considering the heating value of the producer gas. The use of wood chips with a moisture content of 11-34 wt% as a fuel resulted in a very low tar content of 10.6 mg/Nm^3 with a lower heating value of 4.56-4.96 MJ/Nm^3. The temperature profile of the LTB Gasifier observed is relatively low compare to the typical moving bed gasifier. The producer gas obtained could be used directly in an internal combustion engine without observing deposits in the engine after operation. The behavior of the LTB gasifier can be described using a simple thermodynamic equilibrium model that has a good agreement with the model parameters.

Keywords: Biomass Gasification, Gasifier Design, Moving bed Gasifier, Tar reduction process, Low-tar, engine generator.

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1. Introduction

Biomass in the form of wood is the oldest form of energy used to meet a variety of energy needs, including electricity generation [1]. Thermochemical conversion of biomass, including pyrolysis, gasification and combustion, is one of the most promising forms of future energy to provide heat and power. Gasification is the key technology of biomass based power generation in which solid biomass reacts with the gasifying agents to produce gaseous molecules, primarily H_2, CO and CO_2; also tars, methane, water and other constituents at a temperature higher than 700°C [2]. Thermal treatment of biomass is dependent on gasification temperature and gasification agents (air, stream or oxygen) which contributes to producer gas outcome, tar species in the producer gas, and solid products such as char, ash and soot [3].

The use of biomass gasification to generate producer gas is one of the most promising sustainable energy options to integrate small-scale energy generation plants in rural applications by means of off-grid technologies. Tar content in the producer gas is a serious problem if the gas needs to be used directly as a fuel for an internal combustion engine (ICE). The tar content of producer gas leaving a biomass gasifier depends on the design of the gasifier and the type of biomass used as fuel [4]. Theoretically, producer gas with a low tar content can be obtained if a high-temperature zone can be created in the partial oxidation zone where all tar loading pyrolysis gases are forced to reside sufficiently long to undergo in the partial oxidation zone [5]. However, increasing the capacity in the partial oxidation zone always becomes ineffective because of bypassing of tar-loaded pyrolysis gas through relatively cold zones.

Downdraft moving bed gasifier designs produce a producer gas with a low tar content (0.015-3.0 g/Nm^3) that can be coupled directly with ICEs [6] or solid oxide fuel cell applications [7]. The producer gas tar content should be less than 50 mg/Nm^3 to guarantee long lasting performance of the complete gasification system, including an engine-generator system [14]. This requirement can be fulfilled with downdraft gasifiers if properly designed [14]. Therefore, it is of utmost importance to generate a combustible producer gas with as little tar as possible in it. Thus, there is a scope to investigate and reduce tar content level lower than that above-mentioned range. Previous works have been developed in order to design downdraft gasifiers able to increase the efficiency of tar reduction in the producer gas. Bui et al. [8] developed a multi-stage reactor design that separates the flaming-pyrolysis zone from the reduction zone. In that design, the tar vapours generated in the first zone are burned or cracked to simple molecules by high temperature in the second zone, improving the gas quality and conversion efficiency. The minimum content of gravimetric tar obtained with this design was 92 mg/Nm^3. Susanto and Beenackers [5] developed a special type of

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modified co-current moving bed gasifier, in which the pyrolizer, pre-combustor and gasifier are fully integrated in a single reactor and works on the principle of multiple air nozzles or a ring type of oxidation zone with internal recirculation of pyrolysis gas in the oxidation zone; in their experiment a minimum of 48 mg/Nm^3 of tar was obtained [5].

Henriksen et al. [6] examined the influence of the partial oxidation process on the tar destruction and conversion in a TwoStage gasifier. In the TwoStage gasifier the gas from the primary pyrolysis stage is exposed to a partial oxidation before passing a char bed. The partial oxidation of the pyrolysis gas provides heat for the following endothermic char gasification reactions and results in a significant reduction of the tar content of less than 15 mg/Nm^3 [9]. The partial oxidation of the pyrolysis gas and the subsequent char bed conversion are the only two significant tar destruction steps in the TwoStage gasification process and therefore interesting fields to study in order to optimize gas quality and yield [4].

Sarafraz et al. [10] developed a new concept for syngas production with a minimum tar content in which a liquid metal oxide (here copper oxide) is implemented as an oxygen carrier for chemical looping gasification. The influence of the different operating parameters on the syngas production was studied. The results showed an exergy efficiency of the liquid chemical looping gasification of 67% and H_2/CO ratio values as high as 2 suitable for gas to liquid (GTL) processes. Sarafraz et al. [11] studied the relative performance of different potential liquid oxygen carriers within a novel system that can be configured for either chemical looping gasification or combustion. Antimony oxide was found to offer strong potential for high quality syngas production because it has a reasonable oxygen mass ratio for gasification. A sufficiently low operating temperature to be compatible with concentrated solar thermal energy and a propensity to generate methane. Sarafraz et al. [12] conducted a thermodynamic assessment for a new configuration of a supercritical water gasification plant with a water–gas shift reactor. Results showed that the supercritical water to fuel ratio (SCW/C) is a key parameter for determining the quality of syngas (molar ratio of H2:CO) and the carbon conversion reaches 100%, when the SWC/C ratio ranges between two and 2.5 at 500–1000 °C.

The main objective of this work is the design, construction and operation of Low-Tar Biomass (LTB) gasifier to reduce tars in the producer gas. The aim is to identify and characterize the LTB gasifier parameters to optimize its performance. The resulting gas from the LTB gasifier is tested for operation of an internal combustion engine. Furthermore, modeling of the gasifier operation is done using an equilibrium-based model and this model is validated with the experiment data.

1.1 Downdraft Gasifier

A downdraft gasifier has a co-current reactor where gasifying agents enters into the reactor at a certain height below the top [13]. The product gas from the pyrolysis zone flows downward through the oxidation and reduction zone, that's giving the name downdraft [13]. Therefore, all the product gas including volatiles are oxidized or cracked down into gaseous products including CO, H_2, CH_4 and CO_2. The producer gas with high temperature (700-900°C) is drawn off at the bottom of the gasifier and results in almost tar free producer gas [13]. There are two types of downdraft gasifier: throatless and throated. *Fig.* 1 shows the schematic view of a throatless downdraft gasifier with temperature gradient along the height.

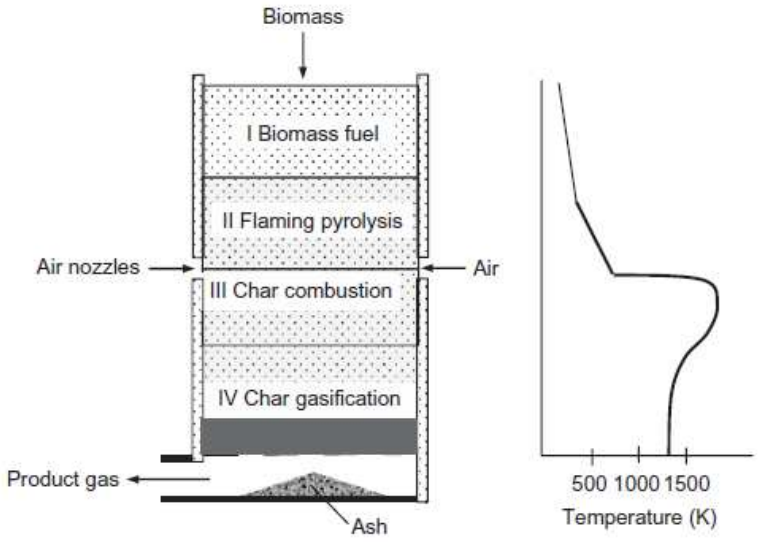


Fig. 1. Schematic view of a throatless downdraft moving bed gasifier (left) with temperature gradient along the height (right) [13].

2. Process Principle

In LTB gasifier design, a separate combustor is installed inside the reactor in the partial oxidation zone. The gasifying agent is fed from the top of the reactor into the combustor by three nozzles, inclined 120 degrees and oriented towards the center of the circle that describes the perimeter of the combustor. This creates a swirl flow that is uniformly distributed on the wall of the combustor and allows to increase the mixing of the gasifying agent with the pyrolysis gas [14]. This mixing gas helps to homogenize the temperature inside the combustor, diminishing the formation of cold areas between the combustor and inside areas in the reactor's partial oxidation zone. Therefore, this novel design mixes the pyrolysis gases with the gasifying air and burns the mixture in an internal separate

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combustor. Additionally, this modified design increases the residence time of the gas inside the combustor and the partial oxidation zone resulting in an increased thermal cracking of tars in the partial oxidation zone, minimizing its passage to the reduction zone, decreasing the tar concentration in the producer gas. Swirl flow produced by nozzles may effectively control the residence time distribution of the gas mixture, which is function of the gasifying agent entry angle [15].

3. Materials and Methods

3.1 Material Used

The LTB gasifier was tested with crushed Danish pine wood pellets and chips. The average size of the wood pellets length was 15-21 mm and diameter of 6 mm. The biggest and smallest size of the wood chips used were 103×39×22 mm and 32×15×6 mm, respectively, while average size of wood chips used were 42×34×6 mm that can be seen in *Fig*. 2. Size and shape of biomass particles will determine the behavior of biomass including bridging and channeling formations and its movement downwards in the moving bed gasifier [16]. Sieving helps to improve biomass quality and homogeneity and avoids small particles and finer dust that may cause a high pressure drop. Also, uniform size and homogeneity of biomass affects the performance of gasifier and increases its efficiency [2]. The elemental composition of biomass was determined using an elemental analyzer (CHNS Elementar Vario GmbH EL III). Carbon, hydrogen and nitrogen were determined using DS/EN ISO 16948 (2015) standard procedure method [17]. Sulphur content was determined using DS/EN ISO 16994 A (2016) method [18]. The heating value (HV) was measured using a bomb calorimeter (IKA C-5000) following the standard procedure DS/EN ISO 18125:2017 [19]. The Moisture content, volatiles and ash composition were determined using the correspondent standard procedures DS/EN ISO 18134-1 (2015), DS/EN ISO 19123 (2015) and DS/EN ISO 18122 (2015), respectively [20].



(a) (b)

Fig. 2. Biomasses used in the gasification process; (a) wood chips, (b) wood pellets

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3.2 Experimental Setup

The gasification setup was built in the facilities that the DTU Chemical Engineering Department has at the DTU Risø Campus in Roskilde, Denmark. *Figure 3* shows the schematic view of the LTB gasifier. It consists of a fuel hopper, nozzle, combustor, ash grate, ash bunker, air inlet system and reactor. *Fig 4* shows different components of the LTB gasifier. The different parts of the LTB gasifier were made of stainless steel (EN 1.4404, AISI 316) having a design capacity of 10 kgh^{-1} for wood chips and 25 kgh^{-1} for wood pellets. The gasifier main body is a cylindrical unit with an inside diameter of 256 mm and a total length of 1480 mm. The fuel hopper has an outside diameter of 270 mm and the biomass is fed through a hopper into the reactor. A converging-diverging nozzle with a diameter of 5 mm is used to inter-connect by 120 degrees the nozzle with the air inlet system for the supply of air into the reactor. The airflow through the nozzle is a rotation flow, such as spiral swirl flow. The combustor has a length of 350 mm and an outside diameter of 114 mm. There are four outside fins around the combustor cylinder at the top that hold the combustor in the gasifier cylinder. The biomass feed is held on an ash grate with circular holes of 5 mm diameter. The remaining ash is collected in the ash bunker. Eight k-type thermocouples are installed in the outside of the gasifier at different heights. Once the different parts are assembled, the insulation material is placed outside the reactor. Finally, the gasifier is placed on a support structure with wheels to easily move it. The safety valve is connected at the top of the gasifier. In order to reduce energy loss due to the heat transfer, the gasifier outside wall is insulated with thermal ceramic material.

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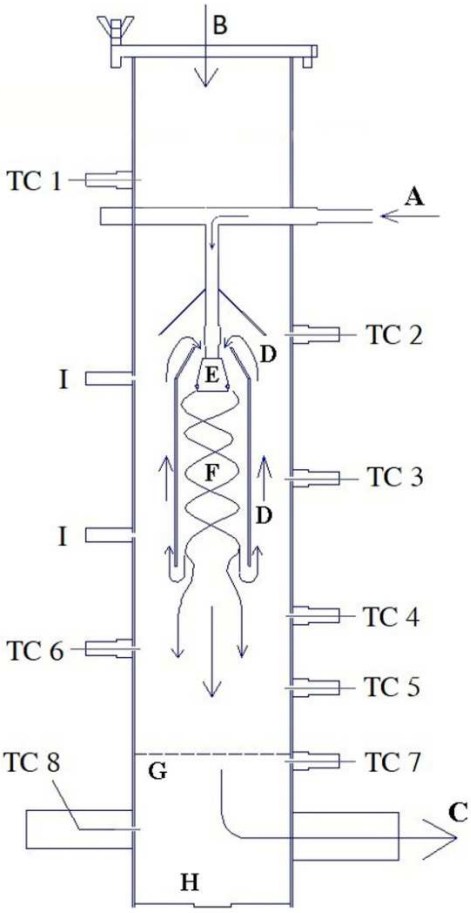


Fig. 3. Schematic diagram of LTB gasifier. A, Air inlet; B, Biomass inlet; C, Producer gas outlet; D, Recycle gas; E, Nozzle; F, Combustor; G, Ash grate; H, Ash bunker; I, Pressure measurement port; TC, thermocouple.



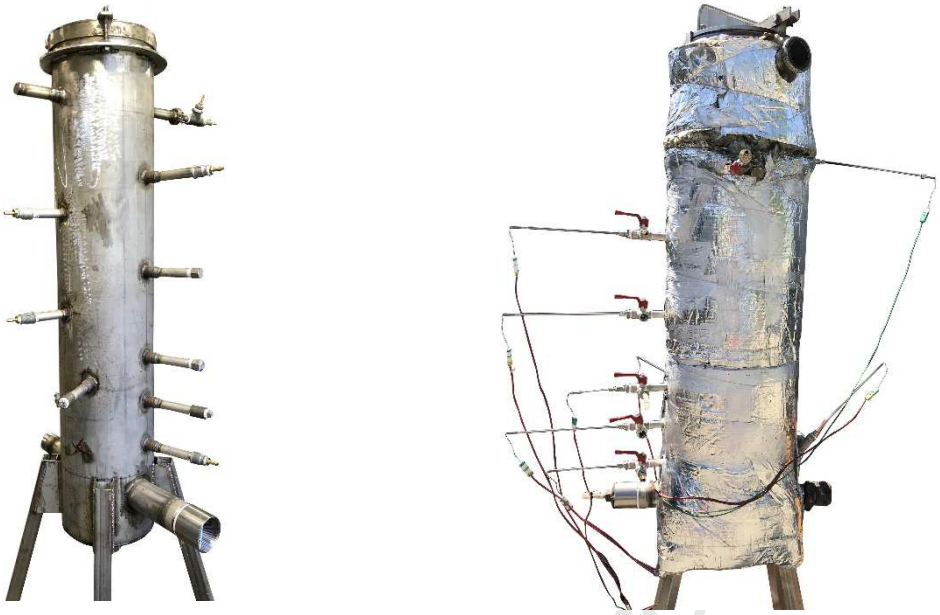
(a)



(b)



(c)



(d) (e)

Fig 4. Different components of the gasifier, namely, (a) Nozzle with accessories, (b) combustor, (c) Ash grate and fuel hopper, (d) Manufacture version of LTB Gasifier, (e) LTB gasifier with insulation

The LTB gasifier lines are heated up to 800-900°C in order to prevent condensation of the producer gas compounds inside the reactor and the measurement device. Gasifier temperatures are measured by eight K-types thermocouples and recorded at intervals of 2s. Airflows are measured continuously by air flow meters. Two liquid column manometers are installed at the combustor and producer gas outlet to check the pressure drop during the testing. Gas composition is measured with a GC analyzer (ABB Automation GmbH, model A02020) at intervals of 1 min. Emissions are measured with a GC analyzer in the exhaust manifold of the engine. Tar analysis is done by solid phase adsorption sampling from the producer gas outlet. All the experimental test data is recorded by a data logger connected a computer. The power generation test was done during the experimental tests. The power generation test consists of a gasification system to the producer gas, which is coupled with a long pipe for cooling the gas and filter for the filtration from particulate matters, and then connected to the diesel generator.

3.3 Process Flow Description

Five tests were run with the LTB gasifier with a duration between 1 h and 5.25 h. Before initiating a test, the gasifier had to achieve thermal stability. The LTB gasifier was placed on a flat surface outdoors and connected to the blower. The blower was used to suck the air out from the gasifier. A LPG cylinder was placed near the reactor. The gas flow through the top of the reactor was turned on to heat up the reactor. A hand-held gas burner (Bernz-O-Matic TS 7000) was introduced to ignite the

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LPG gas to heat up the reactor to 800-900°C. After reaching a stable temperature, the LPG cylinder and the blower were detached from the gasifier. The reduction zone below the combustor was then filled with hot char and fresh cold char was added into the reactor char bed. The whole reactor was filled with biomass. The valve of the air supply system was then turned on to supply the gasifying air into the reactor. After few moments, producer gas started to come out in the form of a thick white smoke through the gas outlet pipe. The profile of reactor temperature and the producer gas composition record was started, when clean producer gas was observed after few minutes. A part of the producer gas was cooled down in the long pipe. Subsequently, it was passed through a filter to get rid of the particulate matters. After filtering, the cool producer gas was set to ducting in the engine for generation of power and electricity.

3.4 SPA Sampling and Analysis

Solid Phase Adsorption (SPA) sampling was used to collect tars contained in the producer gas as shown in *Fig.* 5. From the gas outlet point, condensable producer gas was sampled slowly with SPA tubes from Supelco with an aminopropyl adsorbent. For each SPA sample a needle, a SPA cartridge with adsorbent material and a syringe were assembled. The needle was inserted through a silicon septum seal and into the pipe with the gas at 400°C. After this, the tars adsorbed in the SPA cartridge were extracted by adding acetone (approximately 10 ml) in a glass vial. 1 ml of phenol D6 standard and 1 ml of PAH standard mixture was added to each sample in the vial and the mixture was kept in a refrigerator overnight at 5°C. After that, the vial was taken from the refrigerator, and the filter and tube were separated from the mixture. The mixture without filter and white matter was transferred to the pear-shaped flask. The acetone was evaporated from the mixture by a rotary evaporator at about 60°C and 200 mbar. The evaporated tar was diluted in 1 mL of acetone and then it was transferred into a test tube for the analysis of tar contents. Prepared samples are analyzed in a Hewlett Packard HP 6890 Gas Chromatograph interfaced to a HP 5973 Mass Selective Detector (Agilent, Denmark) and the sample analysis is carried out according to the methodology described in Ref. [9].

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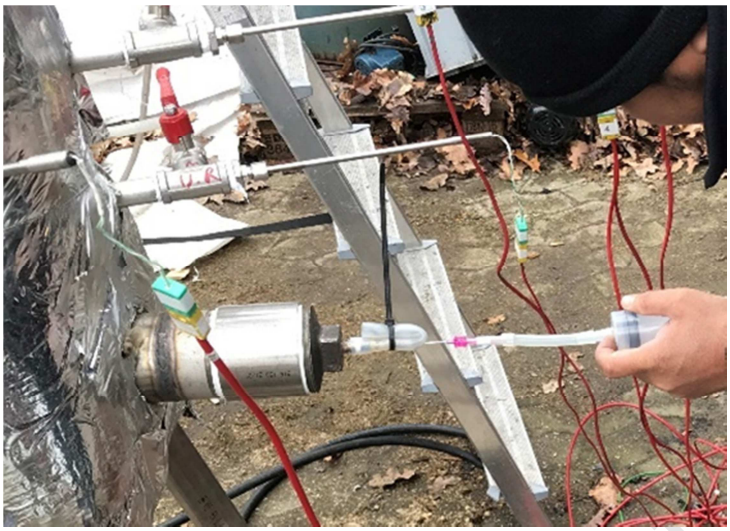


Fig. 5. SPA sample taken from the gas outlet

3.5 Test Consideration

Five tests were carried out for the present study. Initially, tests were planned to be carried out with both wood chips and pellets. Tests 1 and 2 were conducted with wood pellets. The data for the accumulated char and ash could not be recorded in test 1. For this reason, mass and energy balances do not include the results from test 1. After test 2, it was decided to continue the tests only with wood chips. Wood pellets caused operational problems because they disintegrated in the drying zone due to the release of moisture. These disintegrated pellets would react with the liquids (mainly tars) produced in the pyrolysis zone making them sticky and preventing them from moving and falling downwards easily. This situation resulted in a very dense bed that made it very difficult for the air to pass through and resulted in high pressure drops. Therefore, all the analysis is carried out only for wood chips, although the results shown are for both wood pellets and wood chips.

3.6 Thermodynamic Equilibrium Model

A thermodynamic Equilibrium model is simple and allow a practical description of the gasification processes with good approximation, such as those occurring in downdraft gasifiers operates close to equilibrium conditions [21]. Model calculations are independent of the gasifiers design, may be more suitable for process studies on the influence of the most fuel and process parameters. An equilibrium modeling for downdraft gasifiers has been found to be very useful in predicting the behaviors of gasifiers, because it has both pyrolysis and gasification products are forced through the hot oxidation zone, so that equilibrium condition is established after a relatively short time [22]. Thus, the model validation study evaluates the producer gas composition and its main characteristics, such as moisture content, equivalence ratio, producer gas composition, heating value of the producer gas and cold gas efficiency. For this purpose, a thermodynamic equilibrium-

based model built in Engineering Equation Solver (EES) is applied to validate the experimental data with model data [23]. The selected model is suitable for a downdraft moving bed gasifier. The errors in the comparison between the model and experimental values are estimated by the root-mean-square (rms) value for each set of data [23], as given below

$$rms = \sqrt{\frac{\sum_i^n (experiment_i - model_i)^2}{n}} \quad (1)$$

Where n is the number of data set.

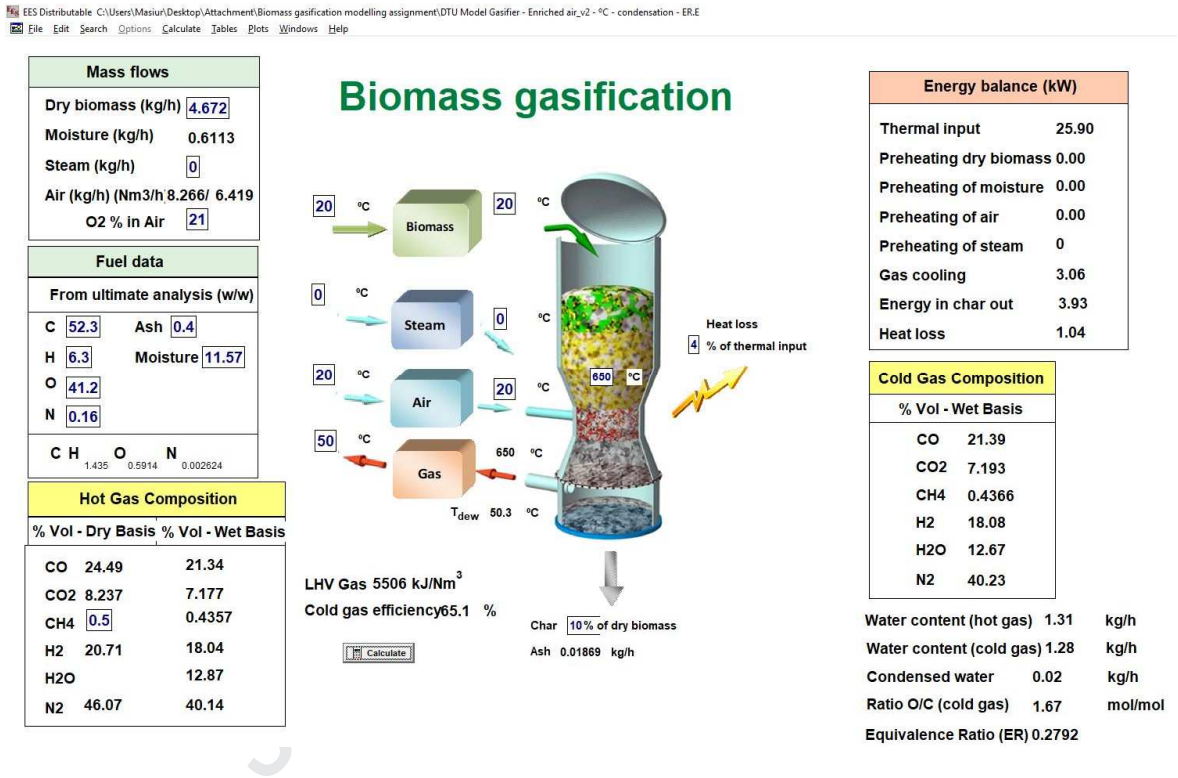


Fig. 6. Engineering Equation Solver (EES) simulator output window for the LTB gasifier operating on wood chips

The numbers inside text boxes of *Fig.* 6 are the inputs of the model. The equilibrium char bed temperature is adjusted between 600 and 700°C and char leaving from the gasifier is given to the model for every test as a char percentage of dry biomass. Heat losses of 4-6% of the energy input are considered for the calculations. The CH_4 percentage leaving the gasifier is also given as an input to the model.

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4. Results and Discussion

4.1 Pyrolysis Results

The properties of the biomass are shown in *Table 1*. The biomass was characterized according to standardized methods. The elemental analysis of biomass shows less amount of oxygen and hydrogen compared to typical values which results in a relatively high heating value of the producer gas. The sulfur content of biomass is exceptionally low, which is an advantage in its utilization for energy conversion. Volatile matter content is high which may result in a higher gasification yield. Ash content was found to be 0.4 (wt%, db). The ash content determined by incineration at 550˚C was higher than that obtained at 815˚C. The discrepancy is caused by the fact that at the higher temperature some of the inorganics contained in the ash fraction can be volatilized. The ash content obtained by incineration at 550˚C was considered more reliable.

Table 1. Biomass characteristics

Parameter	Unit	Results	
		as received	Dry basis
HHV	MJ/kg	19.30	21.20
LHV	MJ/kg	17.82	19.83
Moisture	wt%	9.0	
Volatiles	wt%	77.8	85.5
Ash	wt%	0.3	0.4
Sulphur (S)	wt%	0.009	0.01
Carbon (C)	wt%	47.6	52.3
Hydrogen (H)	wt%	5.8	6.3
Nitrogen (N)	wt%	0.14	0.16
Oxygen (by difference)	wt %	37.15	40.83

4.2 Temperature Profile

Fig. 7 shows a typical behavior of the temperature profile in the gasifier during the testing period, at regular intervals of 2 sec. It is observed that temperature values oscillated in all the bed zone during all the tests, except for the producer gas outlet temperature, which was more stable. This variation can be explained by the biomass movement inside the gasifier during the gasification process. Average partial oxidation temperature observed for wood chips was 968°C which is lower than the typical value of 1100°C reported for this type of gasifier [5][24]. This temperature is slightly higher in test 3 & 4 than in test 5, because the moisture content of the biomass is higher. This higher moisture

content results in a higher temperature and the need of additional energy to evaporate the water and conduct the water gas shift reaction. The producer gas outlet temperature was 676°C which is higher than the ash melting point (550°C) [25]. Thus, this gasifier offers a prospective opportunity for handling biomass with a low-melting ash temperature and this lower temperature results lower heat loss from the systems.

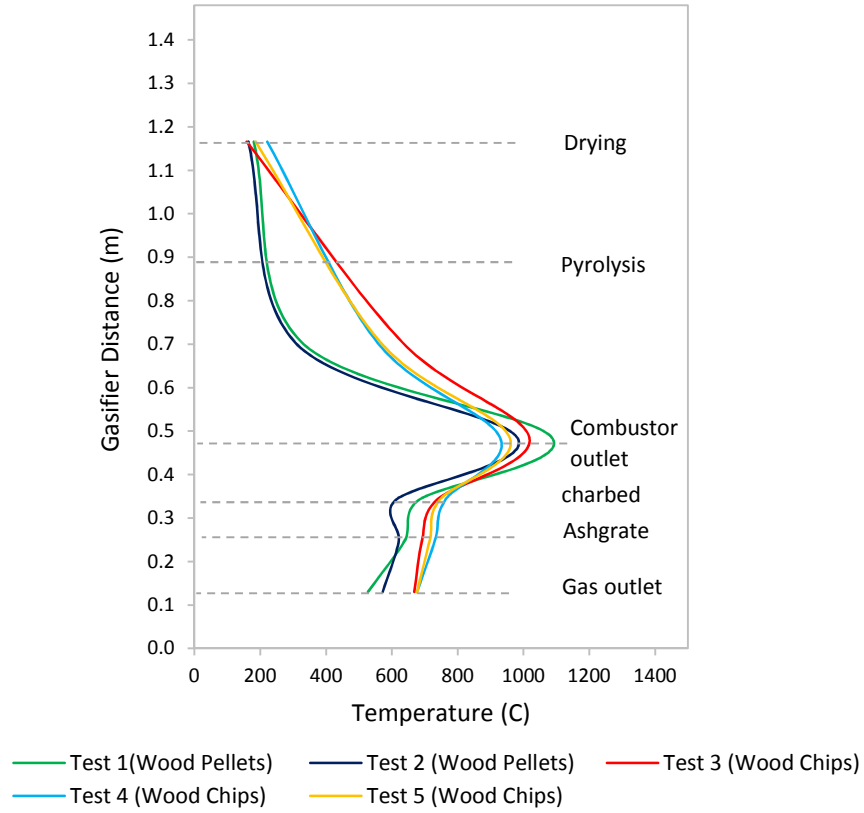


Fig. 7. Temperature profile along the reactor height for all tests using wood chips and pellets

4.3 Gasifier Performance

Gasifier performance is measured in terms of both quality and quantity of the gas produced and its heating value. Here, the amount of biomass converted into producer gas is expressed by cold gas efficiency. *Table 2* shows the gasifier performance for the different tests.

Table 2. Tests results of LTB gasifier

Feed	Unit	Test 2	Test 3	Test 4	Test 5
Biomass type		Wood Pellets	Wood chips	Wood chips	Wood chips
Operating Parameters					
Process time	h	0.92	1.84	5.0	5.25
Moisture content	%	9.5	32.38	34.0	11.57
Biomass feed	Kg	20.85	14.11	43.21	27.74
Air flows	Nm^3	7.41	13.72	44.42	53.09

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	Kg	7.42	16.44	53.74	64.23
Mass Balance					
Inputs					
Biomass	Kg/h	22.75	7.69	8.64	5.28
Char	Kg/h	2.86	0.67	0.13	0.2
Gasifier Air (20^0C, 1 bar)	Kg/h	8.09	10.26	11.38	12.62
Outputs					
Producer Gas	Kg/h	19.15	17.07	19.57	17.52
Wood	Kg/h	4.8	-	-	-
Char	Kg/h	8.54	1.42	0.6	0.52
Ash	Kg/h	0.005	0.141	0.039	0.053
Tar	Kg/h	1.38	0.005	0.0013	0.0005
Mass in/mass out		0.99	1.00	1.00	1.00
LHV of Gas	MJ/Nm^3	5.95	4.57	4.58	4.96
HHV of Gas	MJ/Nm^3	6.54	4.99	5.01	5.29
Gas density	kg/Nm^3	0.955	1.017	1.025	1.038
Dry Gas analysis					
CO	(vol.%)	16.65	17.02	16.32	24.3
H_2		23.77	20.62	19.87	15.85
CO_2		9.81	11.79	12.13	7.11
CH_4		3.54	0.53	1.03	0.49
N_2		46.05	49.9	50.21	52.01
O_2		0.176	0.141	0.44	0.24
Operating Ratios					
O_2/dry biomass		0.14	0.36	0.40	0.55
CO/CO_2		1.7	1.44	1.38	3.42
H_2/CO		1.43	1.21	1.19	0.66
Air/fuel	Nm^3/kg	0.61	0.93	1.01	1.91
Energy Balance					
Inputs					
Wood	MJ	400.12	202.34	604.51	520.13
Char	MJ	89.30	41.89	22.17	35.81
Outputs					
Producer Gas	MJ	84.65	145.24	493.94	439.53
Wood	MJ	92.09			
Char	MJ	291.05	88.77	102.30	88.66
Energy in/energy out		1.05	1.04	1.05	1.05
Heat loss of thermal input	%	4.42	4.18	4.86	4.99
Gasifier Parameters					
Producer Gas Rate	Nm^3/h	20.06	16.78	19.1	16.88
Producer gas yield	Nm^3/kg	1.81	2.09	2.16	3.2
Equivalence Ratio		0.13	0.27	0.29	0.41
Cold gas efficiency	%	60.5	75.4	80.8	91.9
Gasifier capacity	kW	32.9	21.3	24.3	23.3

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A higher moisture content of the biomass in test 3 and 4 resulted in a higher H_2 content in the producer gas. Biomass with high moisture content generates more water vapor in the drying zone that reacts, in the reduction zone, with the char according to steam reforming and water gas reactions to produce more H_2. All the CO formation reactions are endothermic and therefore high temperatures favor CO formation. The lower CO_2 content and higher CO content from producer gas in test 5 are due to the Bouduard reaction. A pressure of nearly 1 atm cannot give a high CH_4 yield and at temperatures above 900°C CH_4 does not exist [26]. The low amount of CH_4 in the producer gas may be due to considerably moderate temperature in the reduction zone.

A control volume is established to calculate the mass balance of the system based on the experimental data obtained and considering that mass in and out is the same. The following mass balance equation is used to determine the amount of producer gas.

$$\frac{dM}{dt} = \dot{m}_{biomass} + \dot{m}_{air} - (\dot{m}_{PG} + \dot{m}_{tars}) \quad (2)$$

Where,

$\dot{m}_{biomass}$ = mass flow rate of biomass consumption, kg/h
$\dot{m}_{air}$ = mass flow rate of air, kg/h
$\dot{m}_{PG}$ = mass flow rate of producer gas, kg/h
$\dot{m}_{tars}$ = mass flow rate of tars, kg/h
$\frac{dM}{dt}$ = Mass of accumulated char and ash in gasifier, kg/h

Assuming a control volume of the gasifier with fixed boundaries, for a steady flow, the energy in and out is the same. The first law of thermodynamics is applied and the following equation is used to calculate the energy balance

$$\frac{dU}{dt} = \sum_i \dot{H}_{in,i} - \sum_j \dot{H}_{out,j} + \dot{Q} + W \quad (3)$$

In which U is the internal energy, $\dot{H}$ is the enthalpy, $\dot{Q}$ is the heat applied to the system and W is the work done on the system. At a constant pressure enthalpy is the internal energy plus the expansion work (H = U + pV), where p is the pressure and V is the volume. In this case, work exerted on the system is assumed to be negligible, only expansion work is considered. The gasifier is equipped with thermal insulation system to avoid heat losses to the surroundings, so the heat transferred to and from the system is assumed to be small, which is denoted by Q_{loss}. For the mass specific enthalpy, h = H/m and considering heat losses, equation (3) becomes

$$\frac{dU}{dt} = \sum_i \dot{m}_{in,i} \times h_{in,i} - \sum_j \dot{m}_{out,j} \times h_{out,j} + \dot{q}_{loss} \quad (4)$$

In which $\dot{m}$ is the mass flow rate (kg/h) and $\dot{q}_{loss}$ is the heat loss.

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The mass and energy balance are used to determine the in-flow and out-flow of materials and energy which shown in *Table 2*. Mass balance involves flow rate of product gas and biomass consumption rate plus air flow rate. It can be observed that, gas production rate is quite similar for all tests for wood chips. The biomass with the lowest moisture content is used in test 5, which results in a higher heating value (HV). In test 3 and 4, biomass has a higher moisture content that requires a considerable amount of sensible heat for evaporation and heating up inside gasifier resulting in the combustion of some combustible gases (H_2, CO and CH_4) that leads to a producer gas with a lower HV. The HV of the producer gas is calculated from the concentration rate of the combustible components. For all the tests run with wood chips, the producer gas HHV was found to be between 4.90 and 5.29 MJ/Nm^3. These values are higher than the experimental results reported in previous works Susanto et al. [5] and Zainal et al. [24] reported 4.7 and 4.85 MJ/Nm^3 respectively for same capacity and same type of moving bed gasifier.

Cold gas efficiency (CGE) is the potential energy output over the energy input [13]. The cold gas efficiency observed for test 5 is the highest (91.9%). This is explained by a lower moisture content of the biomass that increases the LHV of producer gas results in high CGE. The average CGE was found to be 82.7% for wood chips, which is in good agreement with reported results for similar gasifiers by [5] and [24]. The solid fraction removed from the bottom of the gasifier contained mainly char and ash. From the mass balance, a solid removal rate of 0.11 kg/Nm^3 was found for air containing 92% C and 8% char-ash in weight basis, while the total energy loss has to appear 0.21 kW through char-ash generation. Hence, the system gives a satisfactory mass balance.

The calculations from the energy balance in *Table 2* confirmed that heat loss from the system to its surroundings has been estimated to be around 4.5% of the energy input in term of HHV. This very low heat loss is confirmed experimentally from the measurements of the wall temperature of the gasifier during testing period. It is to be noted that the outside surface of the gasifier was well insulated, which minimized heat losses and thus improved the gasifier performance. Heat losses have a significant negative effect on the gasification efficiency.

The CO/CO_2 and H_2/CO ratios are constant for tests 3 and 4, but showed opposite trend for test 5; heating value of the producer gas is a direct consequence of its gas compositions, which mainly depends on its reaction conditions rather than the heating value of biomass. Higher air fuel ratio and low moisture content of biomass which results lower H_2 formation and higher CO formation in test 5 that's get opposite trend.

4.4 Optimization of the Operation Parameters

The Equivalence Ratio (ER) is defined as the ratio of the moles of oxygen supplied to the gasifier to those required for stoichiometric combustion [23]. The air fuel ratio is an important parameter that influences the products of gasification, including gas composition. *Fig.* 8 shows the influence of ER on gas composition and LHV of producer gas. Varying the ER and gasification temperature has the same effect on the producer gas compositions, heating value and gasification efficiency [23]. The CO percentage increases and H_2 percentage decreases when ER increases from 0.27 to 0.41, the CO_2 decreases with the same decreasing trend of H_2. A slight increase of LHV observed in this condition; because higher ER is regulated mainly by high amount of air fuel ratio and low amount of moisture content of biomass which results lower H_2 formation and higher CO formation. CO_2 is producing through the reaction of CO and the available O_2, therefore the exothermic CO shift reaction produces less CO_2 and H_2 at higher temperatures. Higher and same percentage of CO and H_2 observed to the relatively same ER of 0.27 and 0.29; the endothermic water gas reaction observed to these ERs. For this reason, increasing the ER and high temperature results in a higher consumption of char and H_2O to produce high amount of CO and H_2. The steam methane reforming reaction reduces CH_4 formation for all ERs, this reaction is endothermic; therefore, an increase of the temperature benefits the forward reaction to reduce CH_4.

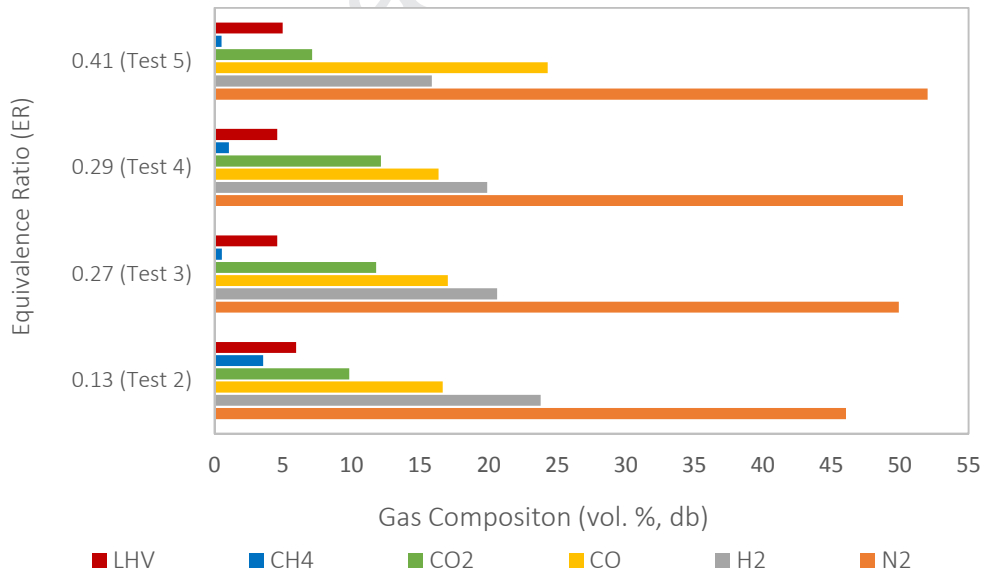


Fig. 8. Effect of the ER on the composition and LHV of producer gas

Fig. 9 shows the variation of the producer gas composition and LHV as a function of the biomass moisture content (MC). Biomass with high MC reduces the quality of the producer gas in term of heating value (HV) [13]. Biomass with more than 30% MC in test 3 and 4 reduces HV due to incomplete pyrolysis processes during gasification [19]. The heat required for complete pyrolysis

process is insufficient because more heat is needed for driving off the moisture during the drying process. Increased levels of MC and the presence of CO produces more H_2 by the water gas shift reaction; consequently, the loss of energy and reduced CO content of the producer gas gives a producer gas with a lower heating value. On the other hand, biomass with less MC produces more a producer gas with a higher heating value. Therefore, biomass with less moisture content is favorable to increase the gas heating value and gasifier efficiency.

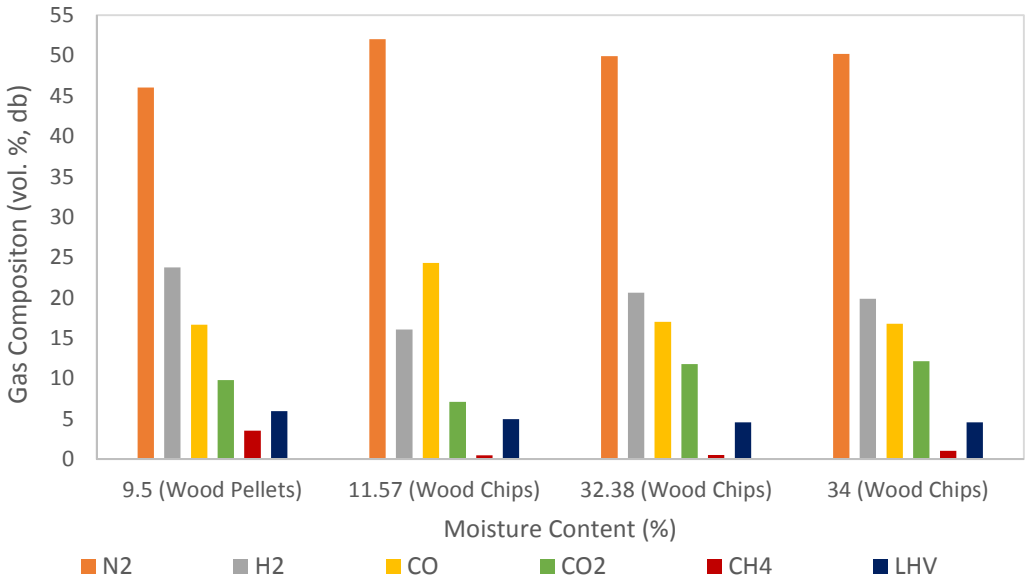


Fig. 9. The effect of biomass MC on the composition of producer gas and LHV

From the above discussion, it can be pointed out that ER plays a significant role in the performance of a LTB gasifier. When considering all three ER for wood chips, the optimum equivalence ratio will be in between 0.29 and 0.41 with respect to the performance of gasifier parameters and this need to be further researched to point out the optimal ER.

4.5 Tar Content

Fig. 10 compares the tar content in the producer gas obtained by several studied Brandt et al. [9], Bhattacharya et al. [27], Walker et al. [28], Reed and Levie [29], Burhenne et al. [30], and Susanto and Beenackers [5]. The gas quality is comparable with that obtained by previous studies and LTB gasifier showed a higher heating value in the same producer gas quality. The tar content analysis showed that average tar content of the producer gas considering all the tests for wood chips was 21.2 mg/Nm^3 and the LTB gasifier can able to reduce tar content of 10.6 mg/Nm^3 for wood chips.

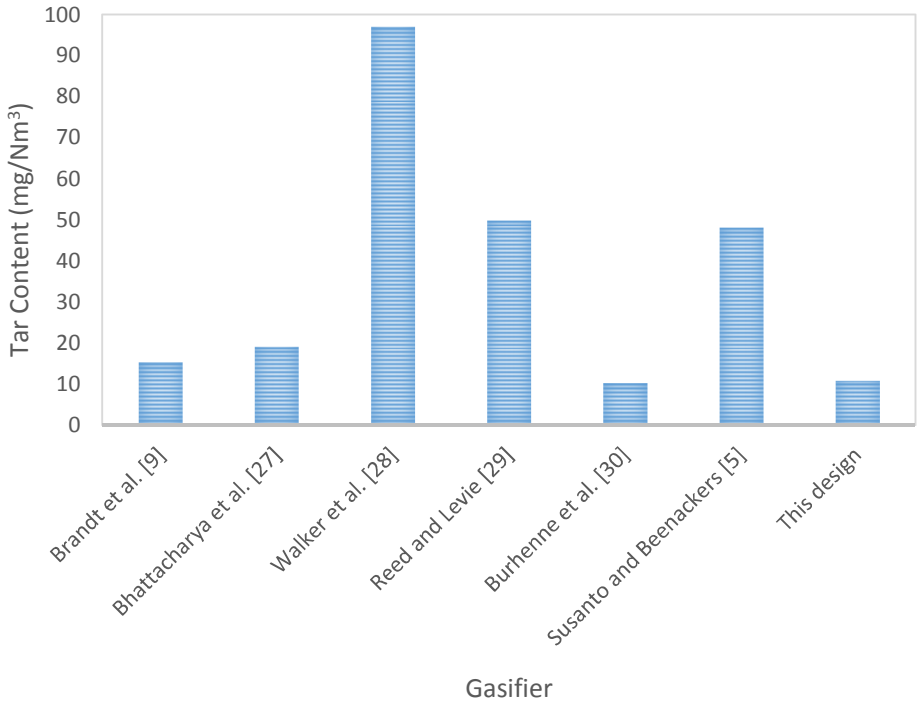


Fig. 10. Comparison between the total tar content obtained by different study and the present study

The tar content analysis shows a great reduction level in tar content of producer gas at the gas outlet for wood chips. The reduction of tar content in the producer gas is achieved by recycling the pyrolysis gas into the gasifying air. Further tar reduction was obtained by using a separate combustor, where gasifier's nozzle created a swirl flow that increased the gas residence time and allowed to increase the mixing of the gasifying agent with the pyrolysis gas and burning the mixture in combustor in a longer time. Finally, extensive tar reduction obtained in the char bed zone where tar from the combustor reacted with the hot char converted into other non-condensable product. As a result, this proved the positive effect of mixing tar with air prior to combustor. Tar content in the producer gas is mainly depends on the design of gasifier and gasifier temperature [13].

4.5.1 Effect of Operating Parameters on Tar Content

Fig. 11 shows the average tar content per mass of fuel as a function of the equivalence ratio and biomass moisture content. It is seen that there is a clear reduction of tar content with increasing ER because the higher ER needs greater amounts of oxygen to react with the volatiles in the flaming pyrolysis zone resulting in lower tar production. Above an ER of 0.27, phenols are nearly all converted, and less tar is formed in the presence of higher temperature [13]. The tar content is higher in biomass having high MC because more heat is needed for driving off the moisture during the drying process and causes incomplete pyrolysis process, and reduction of gasifier's temperature that results in a producer gas with a considerable amount of tar.

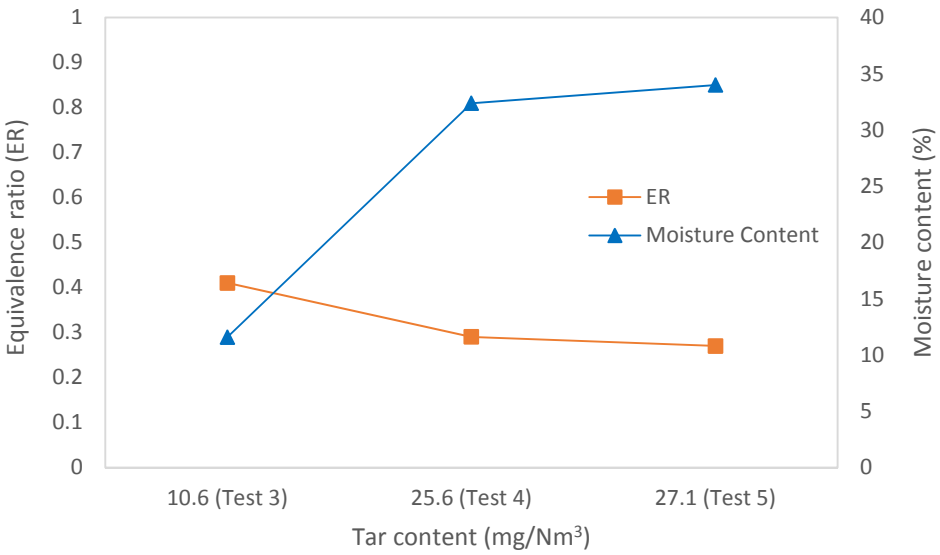


Fig. 11. Effect of ER and MC of wood chips on the tar content of producer gas

4.5.2 Tar Composition

Table 3 shows the composition and quantity of each tar sampled by SPA at the gas outlet and identified by stable isotope GC/MS analysis for wood chips tests. The data of test sample can be found in appendix B. Tar compounds in the producer gas were mainly fluorene and PAHs. It is essential for the efficiency of tar decomposition in the char bed, where tars need to be removed from the producer gas and the remaining tars consists of PAH compounds only and it should be done by the thermal cracking in the partial oxidation zone. When the pyrolysis gases are entering into the partial oxidation zone, where the gases are exposed the temperatures over 900°C thus all the primary tars are break down and the secondary tar increases at the expense of the primary tar. Once the primary tar is nearly destroyed, tertiary tar starts appearing with increasing temperature [13]. Once the temperature is increased to 900°C, the tertiary tar increases light PAH to heavy PAH. In this cases, the tar compounds found in producer gas are light PAH – acenaphthylene, Fluorene and anthracene, and rest of the compounds are found in heavy PAH (without naphthalene) [31].

Table 3. The Composition of tars at a function of the gas outlet.

Name	Test 3				Test 4		Test 5	
	Repetition 1	Repetition 2	Repetition 3	Repetition 4	Repetition 1	Repetition 2	Repetition 1	Repetition 2
Acenaphtene	6.92	6.51	7.36	6.87				3.0
Fluorene	4.62	4.62	13.08	12.48				
Anthracene	4.4	2.16						
Pyrene	4.21	4.9	4.21	3.53	2.3	2.3		
Benz[a]anthracene	4.74	5.51	4.85	5.55				
Benzo[b]fluoranthene		0.7		1.35	9.8	9.8	10.7	
PAH					13.6	13.5		
Methylanthracene								7.52
Total	24.89	24.4	29.5	29.78	25.7	25.6	10.7	10.52
Mean		27.14				25.65		10.61

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SE*	1.41	0.05	0.09

*SE= Standard Error

4.6 Engine and Power Performance

A diesel generator was coupled to the LTB gasifier to investigate engine and power performance with the producer gas obtained. The methodology adopted allows the mixture of air and producer gas into the engine cylinder and allows the gasoline governor to take control on the speed. The gasoline was used only to start the engine. By adjusting the air and gas supply valve, the air fuel ratio is fixed at optimal condition to run the engine smoothly. At this condition, the producer gas was used to start and run the engine for the entire duration of the test. The maximum output capacity of the generator is 3 kW. The generator output power was 1.5 kW continuously and a maximum power output of 2.5 kW was found during the test 5. The power output was tested by connecting a heating device of 2 kW and a lighting device of 0.5 kW. After testing, the cooling pipe, filter, engine carburetor and cylinder head were detached and observed. Neither particulate matters nor ash particles were observed inside the filter. Furthermore, carburetor and the cylinder head appeared to be tar free and clean.

4.7 Emission Measurement

Table 4 shows the emission results in the engine generation system at a power output of 1.5 kWe. Emissions guidelines from biomass gasification plants for gas engine applications suggest the typical standards for emission used in Denmark. Denmark considers 500 mg/m^3 for the CO emission and 550 mg/m^3 with 5% oxygen in the exhaust gas [32]. *Table 4* clearly shows that the CO level is higher and NO_x level is lower than the standard level. The experimental value of CO is comparable with the result found in [32] and the emission of CO is likely same for both experiment. The CO emission is mainly depending on the engine property and the producer gas compositions, and also severe at low power (electrical) output.

Table 4. Emission results in the engine-generator system

Parameters	Value (ppm)	Value (mg/Nm^3)	(vol. %)
CO	9459	6306	
NO	1	1.23	
NO_2	-18	-33.87	
CO			2.46
O_2			9.62
N_2			79.19

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5. Modelling of Experiment Data with the Model Data

Table 5 shows that an equilibrium char bed temperature of 650°C in the model gives the best fit between model and experimental results for gas composition and heating value. This 100°C deviation between model and experimental char bed temperature is considered acceptable. The char bed temperature obtained in the present experiments is lower than the values of 806°C and 850°C reported for wood gasification [5] with a similar gasifier.

Table 5. Comparison of predicted results from modified equilibrium model to experimental data

Test	Dry Biomass	Gasifying Air	Temperature, °C		Heat Loss (%)		Gas Composition (vol %, dry basis)						Heating Value (MJ/Nm3, dry basis)	
			Experimental	Model	Experimental	Model	Experimental	Model	Experimental	Model	Experimental	Model	Experimental	Model
	Kg/h	Nm3/h					CO		CO_2		H_2		LHV	
3	5.20	7.46	736	650	4.18	4	17.0	15.1	11.8	13.5	20.6	19.3	4.57	3.95
4	5.70	8.88	762	650	4.86	4	16.3	17.2	12.1	12.8	19.9	18.3	4.58	4.51
5	4.67	10.11	745	650	4.99	4	24.3	24.5	7.1	8.2	15.9	20.7	4.96	5.51
							0.51		2.03		1.15		0.07	

The model heat loss from the overall gasification process is close to the experimental heat loss in all tests. Heat loss is assumed to be equally distributed among the reduction zone, the oxidation zone and the pyrolysis zone. Heat loss from the combustor was neglected because of the combustor is totally enclosed by the other zones.

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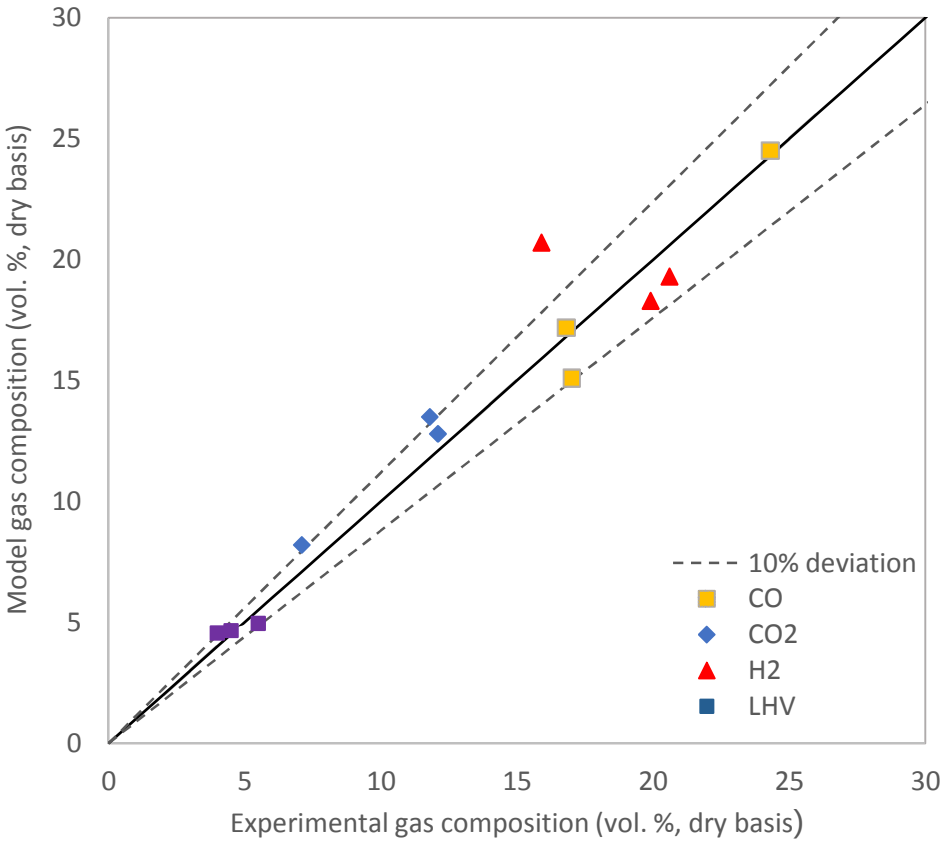


Fig. 12. Comparison of predicted gas compositions results from the equilibrium model with experimental data in test 3, 4 and 5

Fig. 12 shows that the experimental data is in good agreement with the predicted model data. The highest difference corresponds to H_2 content obtained in test 5 with the lowest biomass moisture content. The presented results show that the thermodynamic equilibrium model can be used to evaluate and estimate the performance of the LTB gasifier.

6. Conclusions

A low-Tar biomass (LTB) gasifier has been designed, developed, and evaluated with the aim to achieve a low tar content in the producer gas. A moving bed co-current gasifier with internal recirculation and combustion of pyrolysis gas in the partial oxidation zone with the help of combustor is the basic principle of the LTB gasifier. The LTB gasifier offers excellent opportunities for achieve a tar content of 10.6-29.8 mg/Nm^3 in the producer gas. This value is much lower compared to the tar content observed in a previous analysis for moving bed downdraft gasifiers considering the heating value and makes the producer gas suitable for direct use in internal combustion engines, without further tar removal. Both wood pellets and wood chips were tested, and wood chips is the recommended biomass for the LTB gasifier in order to get almost tar free producer gas and avoid operational problems.

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The heating value of the producer gas is higher than the typical values reported for moving bed downdraft gasifiers. The optimum equivalence ratio lies between 0.29 to 0.41 for the best performance of the LTB gasifier. Using mass and energy balances, the average gasifier capacity and cold gas efficiency is found to be about 23.1 kW and 82.7% for wood chips, whereas 33.1 kW and 60.5% for wood pellets. Average heat loss in term of higher heating value is 4.67% of thermal input for wood chips and 4.42% for wood pellets. Thus, the LTB gasifier performed better compare to a typical moving bed gasifier in terms of heat loss.

The temperature profile of the LTB gasifier observed relatively low compare to typical downdraft moving bed gasifier. The experimental data for LTB gasifier performance is in good agreement with the results provided by the thermodynamic equilibrium model. During the engine generation tests the engine is started and operated completely with the producer gas. The engine operated well on the produced gas, and no deposits are observed in the engine afterwards. Therefore, this gasifier is an excellent technology for thermal and power applications, especially in rural applications.

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Abbreviations

ar	as received
CGE	cold gas efficiency
db	dry basis
DTU	technical university of denmark
EES	engineering equation solver
ER	Equivalence Ratio
GC	gas chromatography
GC/MS	gas chromatography–mass spectrometry
HV	heating value (MJ/Nm^3)
HHV	higher heating value (MJ/Nm^3)
ICE	internal combustion engine
LHV	Lower heating value (MJ/Nm^3)
LPG	liquefied petroleum gas or liquid petroleum gas
LTB	low-tar biomass
MC	moisture content
PAH	polycyclic aromatic hydrocarbons
rms	root-mean-square
SPA	solid phase adsorption
TC	Thermocouple
wb	wet basis

Nomenclature

The symbols used in equations in the text are explained immediately below the equation where the symbols appear.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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